

Program 2

Develop a Multi Stage Dockerfile for Container Orchestration

Project Structure

```
prog2/  
>Dockerfile  
>package.json  
>package-lock.json  
>src/index.js  
>node_modules
```

Step 1 : Create a Dockerfile

```
GNU nano 7.2 Dockerfile *  
#author : Uday R  
#created on: 5th November 2025  
  
# Stage 1: Build Stage  
FROM node:20-alpine AS builder  
  
# Setting working directory  
WORKDIR /app  
  
# Copying package files and installing dependencies  
COPY package.json package-lock.json ./  
RUN npm install  
  
# Copying application source code  
COPY . .  
  
# Build the application (if using a build step, e.g., for React, Next.js)  
RUN npm run build  
  
# Stage 2: Production Stage  
FROM node:20-alpine  
  
# Set working directory  
WORKDIR /app  
  
# Copy only necessary files from build stage  
COPY --from=builder /app/package.json ./  
COPY --from=builder /app/package-lock.json ./  
COPY --from=builder /app/dist ./dist  
COPY --from=builder /app/node_modules ./node_modules  
  
# Expose the application port  
EXPOSE 5000  
  
# Start the application  
CMD ["node", "dist/index.js"]
```

Step 2 : Make a folder src and create a file index.js

```
GNU nano 7.2 index.js  
const express = require('express');  
const app = express();  
const PORT = 5000;  
  
app.get('/', (req, res) => {  
  res.send('Hello from multi-stage Docker!');  
});  
  
app.listen(PORT, () => {  
  console.log(`Server running on port ${PORT}`);  
});
```

Step 3 : Run command - `npm init -y`

This Creates a package-lock.json file.

```
1rv24mc111_uday@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog2$ npm init -y
Wrote to /home/udayraj/lab/prog2/package.json:
```

```
{
  "name": "prog2",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo `Error: no test specified` && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

```
GNU nano 7.2 package.json
{
  "name": "prog2",
  "version": "1.0.0",
  "description": "Multi Stage Dockerfile",
  "main": "build/index.js",
  "scripts": {
    "start": "node dist/index.js",
    "build": "mkdir -p dist && cp -r src/* dist/"
  },
  "keywords": [],
  "author": "Uday R",
  "license": "ISC",
  "type": "commonjs",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

Step 4 : Install node modules - npm install

```
1rv24mc111_uDAY@udayraj-HP-Laptop-15s-eq2xxx:~/lab/prog2$ npm install
```

up to date, audited 70 packages in 911ms

14 packages are looking for funding
run `'npm fund'` for details

found 0 vulnerabilities

Step 5 : Execute Docker build Command - `docker build -t prog2 .`

```
1rv2dc411_u4day@budyajrjw-HP-Laptop-15s-eqe2xxx:~/Lab/prog25$ docker build -t prog2 .
[+] Building 3.4s (15/15) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 824B
=> [internal] load metadata for docker://library/node:28-alpine
=> [auth] library/node:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker://library/node:28-alpine@sha256:6178e78b97f9c335df2814f6764a2d85071aee2af028ffa39a778265435
=> [internal] load build context
=> => transferring context: 1.16KB
=> [internal] load build context
=> CACHED [builder 2/6] WORKDIR /app
=> CACHED [builder 3/6] COPY package.json package-lock.json ./
=> CACHED [builder 4/6] RUN npm install
=> [builder 5/6] COPY . .
=> [builder 6/6] RUN npm run build
=> [stage-1 3/5] COPY --from=builder /app/package.json ./
=> [stage-1 4/5] COPY --from=builder /app/package-lock.json ./
=> [stage-1 5/5] COPY --from=builder /app/dist ./dist
=> exporting to image
=> => exporting layers
=> => writing image sha256:3a2d1d54a9da32c716426c2b7803c25f2a48498c41ef0f5f7fcb3d99d35e
=> => naming to docker://library/prog2
```

```
docker:default      0.0s
                    0.0s
                    2.8s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
                    0.4s
                    0.0s
                    0.0s
                    0.0s
                    0.0s
```

Step 6 : Run the Docker run command

Step 7 : In Web Browser open the following
<http://localhost:5000>

